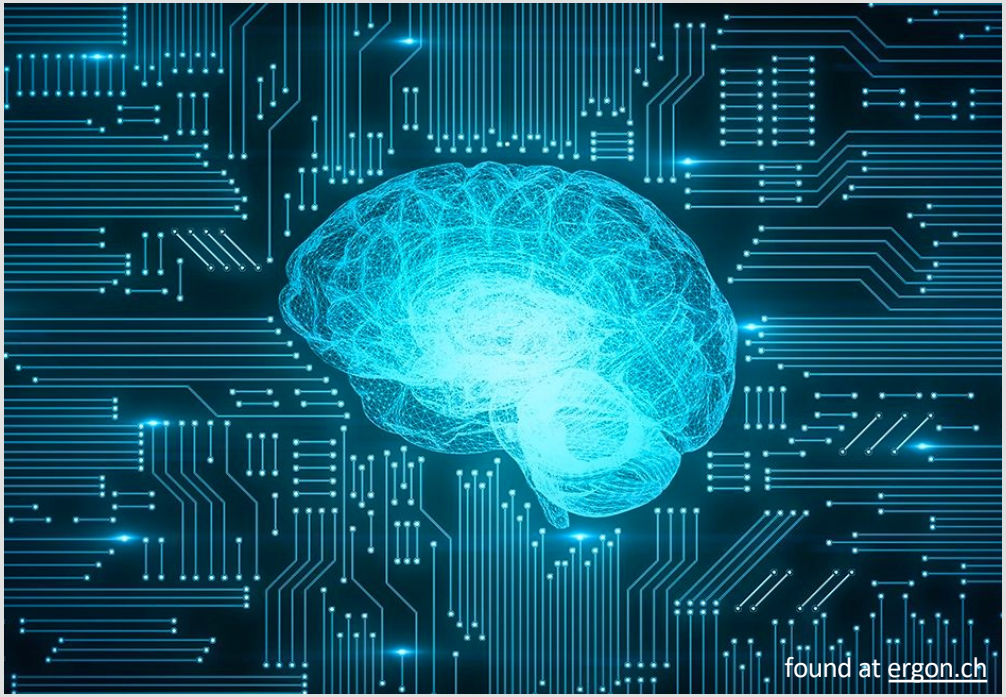
MONASH University
Information Technology

FIT5201 - Machine Learning



Team

Chief Examiner:

- Mario Boley

Lecturers:

- Mario Boley (Weeks 1-6+11-12)
- Bohan Zhuang (Weeks 7-10)

Teaching Associates:

- Bruce Chen, Admin TA (bruce.ying.chen@monash.edu)
- Chris Yun
- Mike Wang
- Haoyu He
- Simon Teshuva
- Shahrzad Behzadi

Point of contact for admin queries
(change lab, enrolment issues, etc.)

Unit Overview

Topic
Fundamental concepts and theory of machine learning

Relationship to other units
- Data Wrangling: pre-process the data before applying ML
- Applied Data Analysis: use ML as blackbox to solve problems
- Deep Learning: similar but explores only one model class, artificial neural networks

Required skills
- Mathematics: probability and linear algebra
- Programming: Python, object orientation, numpy
- Basic Algorithmics: Iterative algorithms

Activities

Type	Description	Assessment Percentage
Lecture/Seminar	2h per week, introduction to topics	unassessed
Lab	2h per week, practical implementation, very close to assignments	unassessed
Moodle quizzes	1 quiz per fortnight, during your own time	9%
Assignment 1	about 4 weeks time, during your own time	25%
Assignment 2	about 4 weeks time, during your own time	16%
Exam		50%

Curriculum / Timetable

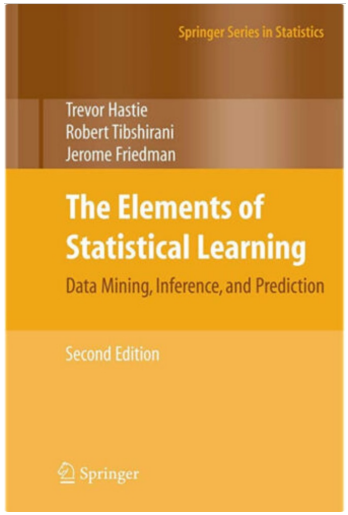
W	Activities	Assessment
1	Module 1 - The Elements of Machine Learning	
2	Module 1 - The Elements of Machine Learning	Online quiz 1 released and assessed
3	Module 2 - Linear Models for Regression	Assignment 1 released
4	Module 2 - Linear Models for Regression	Online quiz 2 released and assessed
5	Module 3 - Linear Models for Classification	
6	Module 3 - Linear Models for Classification	Online quiz 3 released and assessed
7	Module 4 - Latent Variable Models and EM	Assignment 1 due and Assignment 2 released
8	Module 4 - Latent Variable Models and EM	Online quiz 4 released and assessed
9	Module 5 - Neural Networks	
10	Module 5 - Neural Networks	
11	Module 6 - Non-parametric Models	Assignment 2 due
12	Module 6 - Non-parametric Models / Revision	

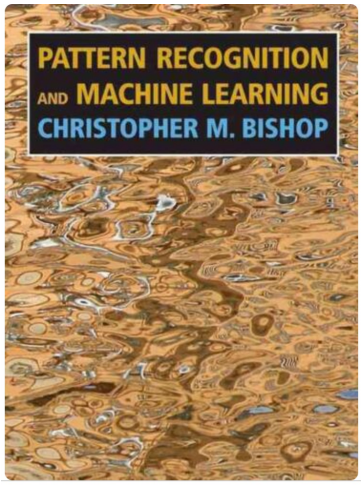
Study Resources

Textbooks

- C Bishop: Pattern Recognition and Machine Learning
- Tibshirani, Hastie, Friedman: The Elements of Statistical Learning

Both well-established authoritative texts on Machine Learning (take time, can be a bit hard to access for beginners)





Moodle Resources

Ask questions

lecture, labs, Ed, consultation